

MA3632 Lecture 4 – Feature Engineering and Dimensionality Reduction

This lecture closes the Data Preparation block. Lectures 1–3 addressed acquiring, cleaning, and transforming data into a consistent numerical form. What remains is the question of *which* variables to present to a learning algorithm, and in what representation. This involves two related but distinct activities. Feature engineering constructs new variables from existing ones, encoding domain knowledge that no algorithm can recover on its own. Dimensionality reduction either selects a subset of the original variables or projects the data onto a lower-dimensional space, addressing the practical and theoretical problems that arise when the number of features is large relative to the number of observations.

The bulk of this lecture is devoted to Principal Component Analysis (PCA), which is the foundational method for dimensionality reduction by projection. It is treated here from first principles via the singular value decomposition, so no prior exposure is assumed.

1 Feature Engineering

Raw features are rarely the best representation of the information available. Feature engineering is the process of constructing new variables from the raw data in order to expose structure that is relevant to the modelling task. It is guided by domain knowledge rather than by any algorithmic procedure, and in practice it is one of the most consequential steps in the pipeline.

1.1 Polynomial and interaction features

A linear model assumes that the expected response is a linear function of the features. If the true relationship is nonlinear, one can extend the feature set with polynomial terms and interaction terms, allowing the model to capture curvature and dependencies between variables while retaining the linear model framework.

Definition 1.1. Given features $X_1, \dots, X_p$, the **degree- d polynomial feature expansion** augments the feature matrix with all monomials $X_1^{k_1} \dots X_p^{k_p}$ satisfying $k_1 + \dots + k_p \leq d$, where $k_j \geq 0$ are non-negative integers. The total number of terms, including the constant term, is $\binom{p+d}{d}$.

For $p = 2$ and $d = 2$, the expansion produces six terms: $1, X_1, X_2, X_1^2, X_1X_2, X_2^2$. The cross-product X_1X_2 is the *interaction term*; it allows the effect of X_1 on the response to depend on the value of X_2 . The number of features grows rapidly with p and d : for $p = 10$ and $d = 3$ the expansion yields $\binom{13}{3} = 286$ features from the original ten. This growth is a special case of the dimensional explosion discussed in Section 2.

1.2 Domain-driven feature construction

The most valuable features are often those that encode prior knowledge about the problem. Some common patterns are the following.

Ratio features. When two variables measure related quantities, their ratio is often more informative than either alone. Body mass index (weight/height²), debt-to-income ratio, and click-through rate (clicks per impression) are all examples where a derived ratio captures the relevant signal more directly than the raw components.

Temporal features. A raw timestamp carries little information for most algorithms, but decomposing it into day of week, hour of day, month, or time elapsed since a reference event often exposes strong periodic or trend structure. Similarly, the difference between two timestamps (time to event, customer tenure) is frequently more relevant than either timestamp individually.

Aggregation features. In datasets with a grouping structure – transactions per customer, readings per sensor, posts per user – aggregating within groups (mean, standard deviation, maximum, count) produces features that describe the group’s behaviour rather than a single observation. These are particularly useful when the unit of prediction is the group rather than the individual record.

Remark 1.2. Feature engineering is where domain expertise enters the pipeline most directly. A well-chosen ratio feature may do more for a linear model’s performance than any amount of algorithmic tuning. Conversely, feeding a large number of poorly motivated constructed features into a model inflates the risk of overfitting and makes the result harder to interpret.

2 The Curse of Dimensionality and Feature Selection

As the number of features p grows, the behaviour of data in $\mathbb{R}^p$ becomes counterintuitive in ways that have direct consequences for learning. This section makes the relevant phenomena precise and then describes filter-based feature selection as a practical remedy.

2.1 The curse of dimensionality

Proposition 2.1. *The volume of the unit ball in $\mathbb{R}^p$ is*

$$V_p = \frac{\pi^{p/2}}{\Gamma(\frac{p}{2} + 1)}.$$

As $p \rightarrow \infty$, $V_p \rightarrow 0$.

Proof. The formula follows by induction on p using the recurrence $V_p = V_{p-2} \cdot 2\pi/p$ with $V_1 = 2$ and $V_2 = \pi$. To see that $V_p \rightarrow 0$, use Stirling’s approximation: for large p , $\Gamma(p/2 + 1) \approx \sqrt{\pi p} (p/2e)^{p/2}$, so

$$V_p \approx \frac{\pi^{p/2}}{\sqrt{\pi p}} \cdot \left(\frac{2e}{p}\right)^{p/2} = \frac{1}{\sqrt{\pi p}} \left(\frac{2\pi e}{p}\right)^{p/2},$$

which tends to zero since $(2\pi e/p)^{p/2} \rightarrow 0$ for $p > 2\pi e$. $\square$

A related high-dimensional phenomenon is that, within a high-dimensional ball, volume becomes increasingly concentrated in a thin shell near its surface: the fraction of a unit ball’s volume lying inside radius $1 - \varepsilon$ is $(1 - \varepsilon)^p \rightarrow 0$ as $p \rightarrow \infty$ for any fixed $\varepsilon \in (0, 1)$, so almost all of the volume lies in the thin outer shell $1 - \varepsilon \leq \|\mathbf{x}\| \leq 1$. For data analysis the consequences are as follows.

Distance concentration. Let $\mathbf{x}_1, \dots, \mathbf{x}_n$ be uniformly distributed on $[0, 1]^p$, with n fixed. As $p \rightarrow \infty$, the ratio of the maximum to the minimum pairwise Euclidean distance satisfies

$$\frac{d_{\max} - d_{\min}}{d_{\min}} \xrightarrow{p} 0,$$

meaning that the relative contrast between the largest and smallest pairwise distances converges to zero in probability, and all points become approximately equidistant. Nearest-neighbour methods, which rely on the notion that nearby points are more similar than distant ones, lose their discriminating power when this ratio approaches zero.

Sample sparsity. To cover $[0, 1]^p$ with a grid of side length ε requires ε^{-p} cells. To have at least one observation per cell in expectation requires $n \geq \varepsilon^{-p}$ observations. For $\varepsilon = 0.1$ and $p = 10$ this is 10^{10} observations – far beyond any realistic dataset. The training data becomes exponentially sparse as p grows, and a model’s ability to generalise from the observed points to unobserved regions deteriorates accordingly.

Remark 2.2. The curse of dimensionality does not affect all algorithms in the same way. Distance-based methods are particularly sensitive, since they rely directly on the distance-concentration phenomenon above. Tree-based and regularised linear methods respond differently to high-dimensional feature spaces – trees split on one feature at a time, and regularisation shrinks the coefficients of uninformative features toward zero – but neither is immune: a large number of irrelevant features still creates more opportunities for spurious splits in a tree, and still degrades performance and increases overfitting risk more generally. Reducing p to the features that genuinely carry information is therefore good practice regardless of the algorithm.

2.2 Filter methods for feature selection

Filter methods assess the relevance of each feature independently of the model, using simple statistical criteria. They are fast and serve well as a first pass to remove obviously uninformative variables.

Variance thresholding. A feature with near-zero variance is approximately constant across all observations and is often, though not always, uninformative – a rare binary indicator, for instance, can have very low variance yet still be highly predictive. Removing near-zero-variance features is nonetheless a common unsupervised screening step, using a threshold τ on the sample variance.

Correlation filtering. If two features X_j and X_k are highly correlated, they carry largely redundant information, and removing one of the pair is a standard redundancy heuristic, though not a guarantee that no useful signal is lost. The standard criterion is to remove one feature from every pair (X_j, X_k) for which $|\hat{\rho}_{jk}| > c$, where $c \in (0, 1)$ is a chosen threshold (commonly 0.9 or 0.95).

Univariate statistical tests. For regression problems the F -statistic for the regression of the target y on each feature X_j individually measures the linear association between X_j and y . For classification problems, mutual information provides a model-free measure of dependence.

Definition 2.3. For a continuous feature X and a discrete target Y , the **mutual information** is

$$I(X; Y) = \sum_y \int p(x, y) \log \frac{p(x, y)}{p(x)p(y)} dx,$$

where $p(x, y)$ is the joint density-mass function and $p(x)$, $p(y)$ are the marginals. (If X is also discrete, the integral is replaced by a sum over x .) $I(X; Y) \geq 0$, with equality if and only if X and Y are independent.

Mutual information captures both linear and nonlinear dependence, unlike the Pearson correlation. Its practical limitation is that estimating it for continuous variables from a finite sample can be statistically noisy and sensitive to the estimation method chosen, particularly when few observations are available.

Remark 2.4. Filter methods treat each feature in isolation and cannot detect features that are individually uninformative but jointly predictive. A feature that carries no signal on its own may become highly relevant in the presence of another feature. Wrapper methods (recursive feature

elimination) and embedded methods (Lasso, tree-based importances) address this by evaluating features in the context of the model, at greater computational cost. Those methods appear in the module from Week 5 onwards as specific model classes are introduced.

3 Principal Component Analysis

Filter methods select a subset of the original features. An alternative approach is to construct an entirely new, lower-dimensional set of features as linear combinations of the originals. Principal Component Analysis (PCA) is the canonical method for doing this; it finds the linear combinations that capture the maximum variance in the data.

3.1 Setup and motivation

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ be the centred data matrix, so that each column has sample mean zero. We seek a unit vector $\mathbf{w}_1 \in \mathbb{R}^p$ such that the projection of the data onto $\mathbf{w}_1$,

$$\mathbf{z}_1 = \mathbf{X}\mathbf{w}_1 \in \mathbb{R}^n,$$

has maximum sample variance. The sample variance of $\mathbf{z}_1$ is

$$\text{Var}(\mathbf{z}_1) = \frac{1}{n-1} \|\mathbf{z}_1\|^2 = \frac{1}{n-1} \mathbf{w}_1^\top \mathbf{X}^\top \mathbf{X} \mathbf{w}_1 = \mathbf{w}_1^\top \mathbf{S} \mathbf{w}_1,$$

where $\mathbf{S} = \mathbf{X}^\top \mathbf{X} / (n-1)$ is the sample covariance matrix. The first principal component direction $\mathbf{w}_1$ is the solution to

$$\max_{\mathbf{w} \in \mathbb{R}^p} \mathbf{w}^\top \mathbf{S} \mathbf{w} \quad \text{subject to} \quad \|\mathbf{w}\| = 1.$$

3.2 Solving the optimisation problem: Lagrange multipliers

Since this is the only point in the module where PCA is introduced, it is worth solving the optimisation problem of Section 3.1 directly, rather than simply asserting which vectors solve it.

Theorem 3.1 (Constrained maximisation of a quadratic form). *Let $\mathbf{S} \in \mathbb{R}^{p \times p}$ be symmetric. A unit vector $\mathbf{w}$ is a stationary point of $\mathbf{w}^\top \mathbf{S} \mathbf{w}$ subject to $\|\mathbf{w}\| = 1$ if and only if $\mathbf{w}$ is an eigenvector of $\mathbf{S}$, in which case the value of the objective equals the corresponding eigenvalue. Consequently*

$$\max_{\|\mathbf{w}\|=1} \mathbf{w}^\top \mathbf{S} \mathbf{w} = \lambda_1,$$

the largest eigenvalue of $\mathbf{S}$, attained at the corresponding eigenvector.

Proof. Introduce a Lagrange multiplier λ for the constraint $\mathbf{w}^\top \mathbf{w} = 1$ and form

$$L(\mathbf{w}, \lambda) = \mathbf{w}^\top \mathbf{S} \mathbf{w} - \lambda(\mathbf{w}^\top \mathbf{w} - 1).$$

Setting $\nabla_{\mathbf{w}} L = 2\mathbf{S}\mathbf{w} - 2\lambda\mathbf{w} = \mathbf{0}$ gives $\mathbf{S}\mathbf{w} = \lambda\mathbf{w}$, so every stationary point is an eigenvector of $\mathbf{S}$ with λ its eigenvalue. At such a point, $\mathbf{w}^\top \mathbf{S} \mathbf{w} = \lambda \mathbf{w}^\top \mathbf{w} = \lambda$.

To identify the global maximum, use the spectral theorem: since $\mathbf{S}$ is symmetric it has an orthonormal eigenbasis $\mathbf{v}_1, \dots, \mathbf{v}_p$ with eigenvalues $\lambda_1 \geq \dots \geq \lambda_p$. Writing any unit vector as $\mathbf{w} = \sum_{i=1}^p c_i \mathbf{v}_i$ with $\sum_i c_i^2 = 1$,

$$\mathbf{w}^\top \mathbf{S} \mathbf{w} = \sum_{i=1}^p \lambda_i c_i^2 \leq \lambda_1 \sum_{i=1}^p c_i^2 = \lambda_1,$$

with equality if and only if $\mathbf{w} = \mathbf{v}_1$ (or any unit vector in the λ_1 -eigenspace, if it has dimension greater than one). $\square$

Remark 3.2. The same argument, applied to the restricted problem of maximising $\mathbf{w}^\top \mathbf{S} \mathbf{w}$ subject to $\|\mathbf{w}\| = 1$ and $\mathbf{w} \perp \mathbf{v}_1, \dots, \mathbf{v}_{k-1}$, shows that the k -th eigenvector $\mathbf{v}_k$ maximises the objective among directions orthogonal to the first $k-1$ eigenvectors, with maximum value λ_k . This is exactly the sequential characterisation of the principal component directions used below.

Example 3.3. Two centred, standardised features have sample covariance matrix

$$\mathbf{S} = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}.$$

The characteristic equation $(1 - \lambda)^2 = 0.8^2$ gives $\lambda_1 = 1.8$ and $\lambda_2 = 0.2$, with eigenvectors $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1)^\top$ and $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(1, -1)^\top$. Geometrically, the covariance ellipse associated with $\mathbf{S}$ is tilted at 45° to the original axes: the first principal component points along the direction in which the two features move together, and it captures $\lambda_1/(\lambda_1 + \lambda_2) = 90\%$ of the total variance, since two strongly correlated features contain a large amount of redundant information that a single direction can summarise. The second component, orthogonal to the first, captures only the 10% of variance reflecting genuine disagreement between the features. This is related to the phenomenon exploited by the correlation filter of Section 2.2: both approaches exploit redundancy among strongly correlated variables, although PCA combines the two features into an optimal linear combination rather than discarding one of them outright.

3.3 The singular value decomposition

Theorem 3.1 characterises the principal component directions abstractly, as eigenvectors of $\mathbf{S}$. For computation, it is more convenient – and, as discussed below, more numerically stable – to work directly with the singular value decomposition (SVD) of the data matrix $\mathbf{X}$ itself.

Theorem 3.4 (Singular Value Decomposition). *Every matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ admits a decomposition*

$$\mathbf{X} = \mathbf{U} \mathbf{D} \mathbf{V}^\top,$$

where $\mathbf{U} \in \mathbb{R}^{n \times r}$ and $\mathbf{V} \in \mathbb{R}^{p \times r}$ have orthonormal columns ($\mathbf{U}^\top \mathbf{U} = \mathbf{V}^\top \mathbf{V} = \mathbf{I}_r$), $r = \min(n, p)$, and $\mathbf{D} = \text{diag}(d_1, \dots, d_r)$ with $d_1 \geq d_2 \geq \dots \geq d_r \geq 0$. This is the reduced (or economy) SVD. The values $d_1, \dots, d_r$ are the singular values of $\mathbf{X}$, the columns of $\mathbf{U}$ are the left singular vectors, and the columns of $\mathbf{V}$ are the right singular vectors. When $n \geq p$ (so $r = p$), $\mathbf{V}$ is square and orthogonal, and this is the case treated below; the case $p > n$, in which the reduced SVD contains $r = n$ singular values (some possibly zero) and hence at most n principal components, is discussed further in Section 3.5.

Proposition 3.5. *For $n \geq p$, the columns of $\mathbf{V}$ are the principal component directions. Specifically, the k -th column $\mathbf{v}_k$ solves*

$$\max_{\mathbf{w} \perp \mathbf{v}_1, \dots, \mathbf{v}_{k-1}, \|\mathbf{w}\|=1} \mathbf{w}^\top \mathbf{S} \mathbf{w},$$

and the corresponding variance explained is $d_k^2/(n-1)$.

Proof. From the SVD, $\mathbf{S} = \mathbf{X}^\top \mathbf{X}/(n-1) = \mathbf{V} \mathbf{D}^2 \mathbf{V}^\top/(n-1)$. This is the eigendecomposition of $\mathbf{S}$: the columns of $\mathbf{V}$ are the eigenvectors and the eigenvalues are $\lambda_k = d_k^2/(n-1)$. By Theorem 3.1 and the remark following it, the eigenvector corresponding to the largest eigenvalue λ_1 maximises $\mathbf{w}^\top \mathbf{S} \mathbf{w}$ subject to $\|\mathbf{w}\| = 1$, and each subsequent eigenvector maximises the quadratic form subject to orthogonality with all previous eigenvectors. The variance of the projection onto $\mathbf{v}_k$ is $\mathbf{v}_k^\top \mathbf{S} \mathbf{v}_k = \lambda_k = d_k^2/(n-1)$. $\square$

Remark 3.6. In practice the principal component directions are computed via the SVD of $\mathbf{X}$ directly, not by forming $\mathbf{S} = \mathbf{X}^\top \mathbf{X} / (n - 1)$ and then eigendecomposing it. Forming $\mathbf{S}$ squares the condition number of the data matrix, which can amplify numerical error substantially when $\mathbf{X}$ is ill-conditioned. Computing the SVD of $\mathbf{X}$ avoids this loss of precision and is generally preferred for numerical stability. In practice, software implementations may choose among several algorithms depending on the shape of the data – `scikit-learn`’s `PCA` class, used in the Week 4 workshop, selects automatically among full, randomised, and covariance-eigendecomposition solvers depending on the dimensions of $\mathbf{X}$ – but the underlying condition-number concern above remains the reason direct SVD is favoured when it is used.

Definition 3.7. For $n \geq p$, the **principal components** of the centred data matrix $\mathbf{X}$ are the projected vectors $\mathbf{z}_k = \mathbf{X}\mathbf{v}_k$ for $k = 1, \dots, p$. Equivalently, from the SVD, $\mathbf{z}_k = d_k \mathbf{u}_k$. The matrix of all principal components is $\mathbf{Z} = \mathbf{X}\mathbf{V} = \mathbf{U}\mathbf{D}$.

Remark 3.8. The principal components $\mathbf{z}_1, \dots, \mathbf{z}_p$ are mutually uncorrelated: $\mathbf{Z}^\top \mathbf{Z} = \mathbf{D}^2$ is diagonal. This decorrelation can be useful for algorithms or analyses that are sensitive to multicollinearity or redundant directions.

3.4 Variance explained and choosing q

In practice one retains only the first $q < p$ principal components, reducing the dimension from p to q . The proportion of total variance explained by the first q components is

$$R_q^2 = \frac{\sum_{k=1}^q d_k^2}{\sum_{k=1}^p d_k^2} = \frac{\sum_{k=1}^q \lambda_k}{\text{tr}(\mathbf{S})},$$

where $\text{tr}(\mathbf{S}) = \sum_{k=1}^p \lambda_k$ is the total variance. The *scree plot* displays λ_k against k and is the standard diagnostic for choosing q : one looks for an “elbow” at which the eigenvalues level off, retaining the components before the elbow. A complementary criterion is to choose the smallest q such that $R_q^2 \geq 0.90$ (or some other pre-specified threshold).

Example 3.9. A dataset of $n = 500$ observations and $p = 20$ financial ratios has sample covariance matrix $\mathbf{S}$ with eigenvalues $\lambda_1 = 4.8$, $\lambda_2 = 2.1$, $\lambda_3 = 1.4$, $\lambda_4 = 0.9$, and the remaining sixteen eigenvalues each approximately 0.2. The total variance is $\text{tr}(\mathbf{S}) = 4.8 + 2.1 + 1.4 + 0.9 + 16 \times 0.2 = 12.4$. The first four components account for approximately $(4.8 + 2.1 + 1.4 + 0.9) / 12.4 \approx 9.2 / 12.4 \approx 74\%$ of the total variance. The scree plot shows a clear elbow after the fourth component, suggesting $q = 4$ is a reasonable choice: a 20-dimensional dataset is reduced to four dimensions while retaining the bulk of its variance structure.

Remark 3.10 (Equivalent characterisation via reconstruction error). Maximising variance is not the only way to characterise the principal components. An equivalent formulation, proved in Exercise 4.2 via the Eckart–Young theorem, is that the first q principal directions minimise the reconstruction error $\|\mathbf{X} - \hat{\mathbf{X}}_q\|_F^2$ over all rank- q approximations $\hat{\mathbf{X}}_q$ of $\mathbf{X}$. The two views – capturing as much variance as possible, and losing as little information as possible under a low-rank approximation – coincide exactly for PCA, which is part of why the method plays such a central role in dimensionality reduction.

3.5 Centring, scaling, and interpretability

PCA is not scale-invariant: features with larger variances can exert disproportionately large influence on the principal components, regardless of whether those features are the most informative. It is

therefore standard practice to standardise all features (zero mean, unit variance) before applying PCA when the features are measured in different units. If the features are all in the same units and their variances carry genuine information, working with the covariance matrix rather than the correlation matrix is more appropriate.

Everything above assumed $n \geq p$. When $p > n$ – as is common in genomics, text, and other high-dimensional settings – the reduced SVD of Theorem 3.3 still exists, now with $r = n$ singular values (some possibly zero) rather than p . For centred data, at most $n - 1$ of these are non-zero (the columns of $\mathbf{X}$ satisfy one linear constraint from centring), so there are at most $\min(n - 1, p)$ non-trivial principal components regardless of how large p is. The maximum-variance and reconstruction-error characterisations of Sections 3.2 and 3.4 continue to hold; the principal component directions remain vectors in the original p -dimensional feature space, and it is only the *number* of non-trivial components that is limited by the rank of the centred data matrix.

A practical limitation of PCA is that the principal components are linear combinations of all p original features and do not correspond to any individual measurable quantity. This makes them difficult to interpret in applied settings where understanding *which* original variable is responsible for a pattern is important. Sparse PCA methods encourage or constrain the loadings $\mathbf{v}_k$ to be sparse, improving interpretability at the cost of a more complex optimisation problem and, depending on the formulation, potentially losing exact orthogonality between components.

Remark 3.11. PCA is an unsupervised method: it maximises variance in the feature space without reference to any target variable y . The directions of maximum variance in X are not necessarily the directions most predictive of y . Supervised alternatives – partial least squares (PLS), linear discriminant analysis (LDA) – find projections that maximise the covariance with y or the separation between classes rather than the marginal variance of X . These will appear in later weeks when the relevant model classes are introduced.

Remark 3.12. Standardised features and their principal components also underlie the **Mahalanobis distance**,

$$d_M(\mathbf{x}) = \sqrt{(\mathbf{x} - \bar{\mathbf{x}})^\top \mathbf{S}^{-1}(\mathbf{x} - \bar{\mathbf{x}})},$$

which measures how far an observation lies from the centre of the data while accounting for the correlation structure captured by $\mathbf{S}$. Expressed in the principal component basis, $d_M(\mathbf{x})$ is simply the Euclidean distance after each coordinate has been rescaled by $1/\sqrt{\lambda_k}$. A Mahalanobis-based approach to flagging unusual observations is therefore a natural extension of the ideas in this lecture, and will reappear when anomaly detection is introduced later in the module.

4 Exercises

Exercise 4.1 (Applied). A dataset has $n = 6$ observations and $p = 3$ features, with centred data matrix

$$\mathbf{X} = \begin{pmatrix} 2 & 1 & 0 \\ -1 & 2 & 1 \\ 3 & 0 & -1 \\ -2 & -1 & 2 \\ 1 & -2 & -1 \\ -3 & 0 & -1 \end{pmatrix}.$$

- Compute the sample covariance matrix $\mathbf{S} = \mathbf{X}^\top \mathbf{X} / (n - 1)$.
- Verify that the columns of $\mathbf{X}$ have sample mean zero.

- (c) Without computing the full eigendecomposition, use the trace and the Frobenius norm of $\mathbf{S}$ to obtain the total variance and an upper bound on the largest eigenvalue.
- (d) The largest eigenvalue of $\mathbf{S}$ is $\lambda_1 = 5.94$ (to two decimal places). What proportion of the total variance does the first principal component explain?

Solution. (a)

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 28 & 0 & -6 \\ 0 & 10 & 2 \\ -6 & 2 & 8 \end{pmatrix}, \quad \mathbf{S} = \frac{1}{5} \begin{pmatrix} 28 & 0 & -6 \\ 0 & 10 & 2 \\ -6 & 2 & 8 \end{pmatrix} = \begin{pmatrix} 5.6 & 0 & -1.2 \\ 0 & 2.0 & 0.4 \\ -1.2 & 0.4 & 1.6 \end{pmatrix}.$$

- (b) The column sums of $\mathbf{X}$ are $2 - 1 + 3 - 2 + 1 - 3 = 0$, $1 + 2 + 0 - 1 - 2 + 0 = 0$, and $0 + 1 - 1 + 2 - 1 - 1 = 0$. All three columns sum to zero, confirming that the data is centred.
- (c) The total variance is $\text{tr}(\mathbf{S}) = 5.6 + 2.0 + 1.6 = 9.2$. Since $\lambda_1 \leq \|\mathbf{S}\|_F$ (the Frobenius norm is an upper bound on the spectral norm for symmetric matrices),

$$\|\mathbf{S}\|_F^2 = 5.6^2 + 2.0^2 + 1.6^2 + 2(0^2 + 1.2^2 + 0.4^2) = 31.36 + 4 + 2.56 + 2(0 + 1.44 + 0.16) = 37.92 + 3.2 = 41.12,$$

so $\lambda_1 \leq \sqrt{41.12} \approx 6.41$.

- (d) The proportion of total variance explained by the first principal component is $\lambda_1/\text{tr}(\mathbf{S}) = 5.94/9.2 \approx 0.646$, so the first component accounts for approximately 64.6% of the total variance. $\square$

Exercise 4.2 (Mathematical). Let $\mathbf{X} \in \mathbb{R}^{n \times p}$, with $n \geq p$, be a centred data matrix with SVD $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$, and let $\mathbf{Z} = \mathbf{U}\mathbf{D}$ be the matrix of principal components.

- (a) Show that $\mathbf{Z}^\top \mathbf{Z} = \mathbf{D}^2$, and deduce that the principal components are mutually uncorrelated.
- (b) Let $\hat{\mathbf{X}}_q = \mathbf{U}_q \mathbf{D}_q \mathbf{V}_q^\top$ be the rank- q truncated SVD, where $\mathbf{U}_q$, $\mathbf{D}_q$, and $\mathbf{V}_q$ retain only the first q columns and diagonal entries. Using the Eckart–Young theorem, show that $\hat{\mathbf{X}}_q$ minimises the Frobenius norm approximation error $\|\mathbf{X} - \mathbf{B}\|_F^2$ over all matrices $\mathbf{B}$ of rank at most q , and compute the minimum value of this error.
- (c) Express the proportion of variance explained R_q^2 in terms of the singular values $d_1, \dots, d_p$ and the approximation error from (b).

Solution. (a) Since $\mathbf{Z} = \mathbf{U}\mathbf{D}$ and $\mathbf{U}^\top \mathbf{U} = \mathbf{I}_p$,

$$\mathbf{Z}^\top \mathbf{Z} = (\mathbf{U}\mathbf{D})^\top (\mathbf{U}\mathbf{D}) = \mathbf{D}\mathbf{U}^\top \mathbf{U}\mathbf{D} = \mathbf{D}^2.$$

The matrix $\mathbf{D}^2$ is diagonal, so the sample covariance matrix of $\mathbf{Z}$ (which is proportional to $\mathbf{Z}^\top \mathbf{Z}$) is diagonal. Off-diagonal entries of the sample covariance are zero, confirming that the principal components are mutually uncorrelated.

- (b) By the Eckart–Young theorem, writing $\mathbf{X} = \sum_{k=1}^p d_k \mathbf{u}_k \mathbf{v}_k^\top$, the minimum of $\|\mathbf{X} - \mathbf{B}\|_F^2$ over matrices $\mathbf{B}$ of rank at most q is attained at $\mathbf{B} = \hat{\mathbf{X}}_q = \sum_{k=1}^q d_k \mathbf{u}_k \mathbf{v}_k^\top$, the rank- q truncated SVD, with minimum value

$$\|\mathbf{X} - \hat{\mathbf{X}}_q\|_F^2 = \sum_{k=q+1}^p d_k^2.$$

This follows since $\|\mathbf{X}\|_F^2 = \sum_{k=1}^p d_k^2$ (using $\|\mathbf{X}\|_F^2 = \text{tr}(\mathbf{X}^\top \mathbf{X}) = \text{tr}(\mathbf{D}^2)$) and the truncated SVD retains exactly the q largest terms of this sum, discarding $\sum_{k=q+1}^p d_k^2$.

(c) From (b), $\|\mathbf{X} - \hat{\mathbf{X}}_q\|_F^2 = \sum_{k=q+1}^p d_k^2$. Since $\|\mathbf{X}\|_F^2 = \sum_{k=1}^p d_k^2 = (n-1) \text{tr}(\mathbf{S})$,

$$R_q^2 = \frac{\sum_{k=1}^q d_k^2}{\sum_{k=1}^p d_k^2} = 1 - \frac{\|\mathbf{X} - \hat{\mathbf{X}}_q\|_F^2}{\|\mathbf{X}\|_F^2}.$$

The proportion of variance explained is therefore one minus the relative squared Frobenius approximation error of the rank- q truncation. Retaining more components reduces the approximation error and increases R_q^2 . □

Exercise 4.3 (Applied). Two standardised features have sample covariance matrix

$$\mathbf{S} = \begin{pmatrix} 1 & -0.6 \\ -0.6 & 1 \end{pmatrix}.$$

- (a) Find the eigenvalues of $\mathbf{S}$ and the corresponding unit eigenvectors.
- (b) What proportion of the total variance is explained by the first principal component?
- (c) Describe the orientation of the first principal component relative to the two original axes, and explain why it makes sense given the sign of the off-diagonal entry of $\mathbf{S}$.

Solution. (a) The characteristic equation is $(1 - \lambda)^2 - 0.36 = 0$, so $\lambda = 1 \pm 0.6$, giving $\lambda_1 = 1.6$ and $\lambda_2 = 0.4$. For $\lambda_1 = 1.6$, $(\mathbf{S} - 1.6\mathbf{I})\mathbf{v} = \mathbf{0}$ gives $-0.6v_1 - 0.6v_2 = 0$, so $v_1 = -v_2$ and the normalised eigenvector is $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, -1)^\top$. Orthogonality gives $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(1, 1)^\top$ for $\lambda_2 = 0.4$.

- (b) $\lambda_1/(\lambda_1 + \lambda_2) = 1.6/2.0 = 0.80$, so the first component explains 80% of the total variance.
- (c) The first principal component lies along $(1, -1)$, at 45° to the axes but tilted the opposite way to the worked example in Section 3.2: since the two features are negatively correlated, they tend to move in opposite directions, and the direction of maximum joint variability is the one in which one feature increases while the other decreases. □

Further Reading

The singular value decomposition derivation of PCA given here draws on the following.

Deisenroth, M. P., Faisal, A. A., and Ong, C. S. (2020). *Mathematics for Machine Learning*. Cambridge University Press. Chapter 10 covers both derivations used in this lecture: the Lagrangian, maximum-variance route of Section 3.2, and the projection-error formulation underlying the reconstruction-error remark of Section 3.4. It also discusses computing PCA via the SVD rather than by eigendecomposing the covariance matrix directly, as in the remark of Section 3.3.

Lindholm, A., Wahlström, N., Lindsten, F., and Schön, T. B. (2022). *Machine Learning: A First Course for Engineers and Scientists*. Cambridge University Press. Chapter 10, on generative models and learning from unlabeled data, treats PCA and related dimensionality-reduction material at a level consistent with the rest of this module, and is a good source for further worked examples of the type given in Section 3.2 and Section 3.3.

James, G., Witten, D., Hastie, T., Tibshirani, R., and Taylor, J. (2023). *An Introduction to Statistical Learning with Applications in Python*. Springer. Chapter 12 (Unsupervised Learning), Section 12.2, covers PCA from an applied perspective, including scree plots and the choice of the number of components, and is a useful companion to the Week 4 workshop.

References

- Deisenroth, M.P., Faisal, A.A. and Ong, C.S., 2020. *Mathematics for Machine Learning*. Cambridge University Press.
- James, G., Witten, D., Hastie, T., Tibshirani, R. and Taylor, J., 2023. *An Introduction to Statistical Learning with Applications in Python*. Springer.
- Lindholm, A., Wahlström, N., Lindsten, F. and Schön, T.B., 2022. *Machine Learning: A First Course for Engineers and Scientists*. Cambridge University Press.